



Course Title: Cryptography and Network Security (EE-426 / CE-442)

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Assignment No. 03 SOL

Release Date: April 15, 2026

Due by: April 24, 2026 11:59 PM

Total points: 100

Points obtained:

Student Name:	Student ID:	Section:
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Purpose:

The purpose of this assignment is to help you evaluate message authentication schemes, key management schemes in addition to the application of cryptography to network security technology.

Instructions:

1. This assignment can be done individually or in pair.
2. All questions should be answered in **black ink only**. (Extra sheets can be used)
3. Scan your answer sheet and upload it on LMS before the due date.

Grading Criteria:

1. Your assignments will be checked by instructor/TA.
2. You can also be asked to give a viva where you will be judged whether you understood the question yourself or not. If you are unable to correctly answer the question you have attempted right, you may lose your marks.
3. Zero will be given if the assignment is found to be plagiarized.
4. Untidy work will result in a reduction of your points.

Late submission penalty:

- 1-day late submission – 20% deduction of the maximum allowable marks
- 2-days late submission – 40% deduction of the maximum allowable marks
- No submission will be accepted after two days of the original deadline

CLO Assessment:

This assignment assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	Apply the concepts from number theory and algebraic structures in understanding the design of cryptographic algorithms.	
CLO 2	Evaluate the suitability of message integrity, message authentication, and key management methods when applying in a given security scenario.	✓
CLO 3	Explain the setup, algorithms, and security issues in existing symmetric-key and asymmetric-key cryptosystems.	
CLO 4	Apply cryptography-based network security technologies in the design of networked information systems.	✓

P#	Questions	Pts
1	Using the RSA scheme, let $p = 809$, $q = 751$, and $d = 23$. Calculate the public key e. Then: a. Sign and verify a message with $M_1 = 150$. Call the signature S_1. b. Sign and verify a message with $M_2 = 75$. Call the signature S_2. c. Show that if $M = M_1 \times M_2 = 11250$, then $S = S_1 \times S_2$. (Note: This is an example of a multiplicative attack on RSA Signature) Answer: Values of p and q are given in the question, $n = p \times q = 809 \times 751 = 607559$ $\phi(n) = (p-1) \times (q-1) = (809 - 1) \times (751 - 1) = 606000$. Since $d = 23$, we have $e = d^{-1} \bmod \phi(n) = 158087$ (a) $S_1 = M_1^d \bmod n = 150^{23} \bmod 607559 = 129525$ $M_1 = S_1^e \bmod n = 129525^{158087} \bmod 607559 = \mathbf{150}$ (b) $S_2 = M_2^d \bmod n = 75^{23} \bmod 607559 = 455282$ $M_2 = S_2^e \bmod n = 455282^{158087} \bmod 607559 = \mathbf{75}$ (c) If $M = M_1 \times M_2 = 11250$, then we have: $S = M^d \bmod n = 11250^{23} \bmod 607559 = \mathbf{116951}$ $S = (S_1 \times S_2) \bmod n = (129525 \times 455282) \bmod 607559 = 58970401050 \bmod 607559 = \mathbf{116951}$	10
2	Using the ElGamal scheme, let $p = 881$ and $d = 700$. Find values for e_1 and e_2. Choose $r = 17$. Find the value of S_1 and S_2 if $M = 200$.	10

	Answer: Values of p and d are given in the question, $p = 881$ $d = 700$. We choose $e_1 = 3$ (primitive root of 881). Then $e_2 = e_1^d \bmod p = 471$. $r = 17$ $S_1 = e_1^r \bmod p = 3^{17} \bmod 881 = \mathbf{540}$ $S_2 = (M - d \times S_1) r^{-1} \bmod (p-1) = (200 - 700 \times 540) 17^{-1} \bmod 880 = (200 - 700 \times 540) 17^{-1} \bmod 880 = 600 \times 673 \bmod 880 = \mathbf{760}$ where $17^{-1} \bmod 880 = 673$ Verification (Optional): We can verify the signature because V1 is congruent to V2. $V_1 = e_1^M \bmod p = 3^{200} \bmod 881 = \mathbf{738}$ $V_2 = e_2^{S_1} \times S_1^{S_2} \bmod p = 471^{540} \times 540^{760} \bmod 881 = 90 \times 713 \bmod 881 = \mathbf{738}$	
3	Using the DSS scheme, let $q = 59$, $p = 709$, and $d = 14$. Find values for e_1 and e_2. Choose $r = 13$. a. Find the value of S_1 and S_2 if $h(M) = 100$. Verify the signature. b. Which digital signature computation is faster than RSA signature computation for the same value of p. Answer: Values given in the question: $q = 59$, $p = 709$, $d = 14$, $r = 13$ and $h(M) = 100$. We choose $e_0 = 2$ (primitive element in Z_p). Then we have $e_1 = e_0^{(p-1)/q} \bmod p = 551$ $e_2 = e_1^d \bmod p = 399$. (a) The following shows the calculation for S_1 and S_2 at the sender site and the calculation of V at the receiver site. $S_1 = (e_1^r \bmod p) \bmod q = \mathbf{48}$	10

	$S_2 = (h(M) + d \times S_1) r^{-1} \bmod q = ((100 + 14 \times 48) \times 50) \bmod 59 = \mathbf{14}$ Verification: $S_1 = 48$ as computed earlier. $S_2^{-1} \bmod q = 38$ $V = (e_1^{h(M) \times 38} \times e_2^{S_1 \times 38} \bmod p) \bmod q = \mathbf{48}$ (b) Computation of DSS signatures is faster than computation of RSA signatures when using the same p.	
4	Describe the steps which are used to sign and verify a digital signature using Elliptic Curve Digital Signature Algorithm. Answer: ♦ Setup (Public parameters) • Elliptic curve E over finite field • Base point G of order n ♦ Key Generation 1. Choose private key: $d \in [1, n - 1]$ 2. Compute public key: $Q = dG$ ♦ Signing a message M 1. Compute hash: $e = H(M)$ 2. Choose random $k \in [1, n - 1]$ 3. Compute point: $(kG) = (x_1, y_1)$ 4. Compute: $r = x_1 \bmod n$ If $r = 0$, choose new k 5. Compute: $s = k^{-1}(e + dr) \bmod n$ If $s = 0$, choose new k 6. Signature is: (r, s)	10

	4	16	1	628	394384 mod n = 6117	394384 mod n = 6117	
	5	25	0	5	25	25	
	<i>Hint: The values of the last two columns should be the same if Alice is honest or has correctly guessed the value of c.</i>						
6	In the Diffie-Hellman key exchange protocol, suppose that $g = 9$, $p = 23$, $x = 4$, and $y = 3$. a. What is the value of the symmetric key K ? b. What is the value of R_1 and R_2 ? Answer: (a) $K = g^{xy} \bmod p = 9^{4 \times 3} \bmod 23 = 9$. (b) $R_1 = g^x \bmod p = 9^4 \bmod 23 = 6$ $R_2 = g^y \bmod p = 9^3 \bmod 23 = 16$. Note that: $K = R_2^x \bmod 23 = 16^4 \bmod 23 = R_1^y \bmod 23 = 6^3 \bmod 23 = 9$.						10
7	With reference to IPSec Protocol, draw a table to compare and contrast the security services provided by AH and ESP protocols. Briefly discuss each service provided by these two protocols. Hint: Refer Table 18.1 in the reference text. Answer: Services Provided by IPSec: The two protocols, AH and ESP, can provide several security services for packets at the network layer. Table 1 shows the list of services available for each protocol. Table 1. IPSec services						30

<i>Services</i>	<i>AH</i>	<i>ESP</i>
Access control	yes	yes
Message authentication (message integrity)	yes	yes
Entity authentication (data source authentication)	yes	yes
Confidentiality	no	yes
Replay attack protection	yes	yes

Access Control

IPSec provides access control indirectly using a Security Association Database (SAD). When a packet arrives at a destination, and there is no Security Association already established for this packet; the packet is discarded.

Message Integrity

Message integrity is preserved in both AH and ESP. A digest of data is created and sent by the sender to be checked by the receiver.

Entity Authentication

The Security Association and the keyed-hash digest of the data sent by the sender authenticate the sender of the data in both AH and ESP.

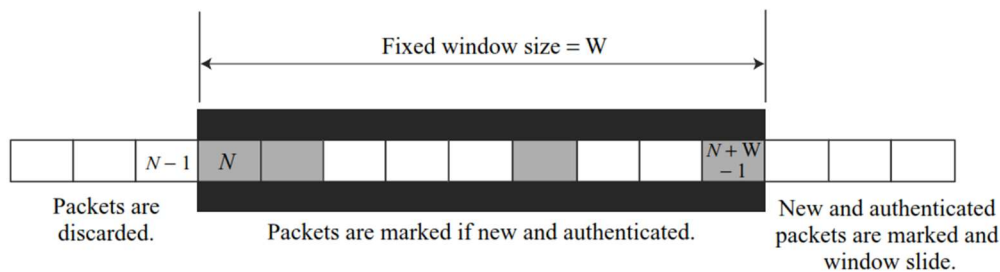
Confidentiality

The encryption of the message in ESP provides confidentiality. AH, however, does not provide confidentiality. If confidentiality is needed, one should use ESP instead of AH.

Replay Attack Protection

In both protocols, the replay attack is prevented by using sequence numbers and a sliding receiver window. Each IPSec header contains a unique sequence number when the Security Association is established. The number starts from 0 and increases until the value reaches $2^{32} - 1$ (the size of the sequence number field is 32 bits). When the sequence number reaches the maximum, it is reset to 0 and, at the same time, the old Security Association (see the next section) is deleted and a new one is established. To prevent processing duplicate packets, IPSec mandates the use of a fixed-size window at the receiver. The size of the window is determined by the receiver with a default value of 64. Figure 18.9 shows a replay window. The window is of a fixed size, W . The shaded packets signify received packets that have been checked and authenticated.

Figure 18.9 *Replay window*



When a packet arrives at the receiver, one of three things can happen, depending on the value of the sequence number.

1. The sequence number of the packet is less than N . This puts the packet to the left of the window. In this case, the packet is discarded. It is either a duplicate or its arrival time has expired.

2. The sequence number of the packet is between N and $(N + W - 1)$, inclusive. This puts the packet inside the window. In this case, if the packet is new (not marked) and it passes the authentication test, the sequence number is marked and the packet is accepted. Otherwise, it is discarded.

3. The sequence number of the packet is greater than $(N + W - 1)$. This puts the packet to the right of the window. In this case, if the packet is authenticated, the corresponding sequence number is marked and the window slides to the right to cover the newly marked sequence number. Otherwise, the packet is discarded. Note that it may happen that a packet arrives with a sequence number much larger than $(N + W)$ (very far from the right edge of the window). In this case, the sliding of the window may cause many unmarked numbers to fall to the left of the window. These packets, when they arrive, will never be accepted; their time has expired. For example, in Figure 18.9, if a packet arrives with sequence number $(N + W + 2)$, the window slides and the left edge will be at the beginning of $(N + 3)$. This means the sequence number $(N + 2)$ is now out of the window. If a packet arrives with this sequence number, it will be discarded.